B21 Retrieval-augmented generation

Chunking, retrieval with embeddings and keywords, and grounded answers with checked citations.

B21-L1 Chunk documents for retrieval

Split on structure, then by size with overlap. Keep every word, and keep the metadata that citations need.

```
words = text.split()
step = size - overlap
starts = range(0, max(len(words) - overlap, 1), step)
chunks = [" ".join(words[i:i + size]) for i in starts]
```

The three steps of RAG	Retrieve relevant chunks, put them in the prompt, and answer from them alone with citations.
Sliding window starts	range(0, max(len(words) - overlap, 1), size - overlap). Every word covered, no window of repeats.
Chunk size trade-off	Small chunks match precisely but lose context. Large chunks keep context but blur the match. Test sizes on an evaluation set.
Metadata on every chunk	Document ID and version, section, position and a stable chunk ID, for citations, filters and re-indexing.

B21-L2 Retrieve with embeddings, keywords and rank fusion

Filter, score by cosine similarity, add a keyword ranking, fuse with RRF, and report when nothing matches well.

```
scores = chunk_vecs @ query_vec
top = np.argsort(-scores)[:k]
```

Top k by cosine similarity with NumPy	Divide dot products by both lengths, then np.argsort(-scores)[:k]. Guard zero lengths.
Reciprocal rank fusion	Each ranking gives a chunk 1 / (k + rank), ranks from 1, summed across rankings. k = 60 in the original paper.
Why add keyword search	Embeddings blur exact tokens such as wave codes and variable names. Keyword ranking matches them.
Before ranking, and after	Filter to chunks the question may use. If the best score is low, report that nothing matched.

B21-L3 Grounded answers, context budgets and citation checks

Tag each source with its ID, fill the budget in rank order without cutting chunks, and check every citation against what you sent.

```
<source id="weighting#2">Weights are capped at 5 after raking.</source>
<question>What is the weight cap?</question>
```

```
cited = re.findall(r"\[([^\]]+)\]", answer)
unknown = sorted(set(cited) - supplied_ids)
```

Rules for a grounded answer	Answer only from the sources, cite [id] after each claim, use the fixed refusal when the sources lack the answer, and treat source text as data.
Filling the context budget	Add tagged chunks in rank order, check used + n before adding, and stop at the first overflow. Never cut a chunk.

Counting tokens in anthropic 2.4.0	client.messages.count_tokens(model=..., system=..., messages=...).input_tokens.
Citation check	re.findall(r"[([^]]+)]", answer). Flag IDs you never sent, and answers without a citation unless they are the refusal.